

The Effect of Perceived Challenges and Skills on the Quality of Subjective Experience

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ABSTRACT This article investigates the effects that perceived challenges and skills in activities have on the quality of everyday life experience. Based on flow theory it was predicted that quality of daily experience would depend on the challenge experienced and skill required in specific situations, as well as on the balance between challenge and skill. The Experience Sampling Method (ESM) was used on a sample of 208 talented adolescents to measure daily variations in four dimensions of experience (concentration, wish to do the activity, involvement, and happiness) in four contexts (in school, with relatives, with friends, and in solitude). The four dimensions of experience were regressed on the predictors challenges, skills, and their absolute difference expressing the balance/imbalance of challenges and skills. Hierarchical linear modeling, explained in detail herein, was conducted on a 1-week sample of experiences. Findings confirm the prediction of flow theory that the balance of challenges and skills has a positive and independent effect on the quality of experience. Yet some differences of parameter estimates were found between dimensions of experience and between social contexts. These heterogeneities call for a further improvement of the flow model.

In the past 20 years, psychologists have developed a keen interest in the study of subjective experience and are realizing that the quality of ex-

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perience is a fundamental part of human psychology for three complementary reasons. First, the quality of life is an intrinsically subjective, experiential construct (Argyle, 1986; Csikszentmihalyi, 1990; Myers, 1992; Strack & Carver, 1987; Strack, Argyle, & Schwartz, 1991). Second, positive subjective disposition is predictive of better health (Levy, Lee, Bagley, & Lippman, 1988; Peterson, Seligman, & Vaillant, 1988; Scheier & Carver, 1987). Third, it appears that subjective experience is a relevant intermediate variable that may mediate the effect of constitutional and environmental characteristics (see review by Singer & Kolligian, 1987).

In 1991 an entire issue of the *Journal of Personality* was dedicated to the study of experience. That issue highlighted problems in the development of both theory and analytical methods. In this article, we address both classes of problems, with a test of predictions from flow theory, a theory of subjective experience, and the presentation of an analytical methodology for handling streams of experiential data, i.e., random coefficients regression or hierarchical linear modeling. Our test is conducted on a sample of talented adolescents.

There are many theoretical approaches to the study of experience, and several attempts have been made to integrate the different theories (e.g., Eckblad, 1981; McAdams, 1990; Singer & Kolligian, 1987). Historically, subjective experience has been considered in terms of curiosity and the exploratory behaviors of rats (e.g., Dashiell, 1925), and in terms of "intrinsic motivation" and "stimulus seeking" (e.g., Berlyne, 1950; Harlow, Harlow, & Meyer, 1950; Hebb, 1949), with a focus on the experiential states in which exploratory and intrinsically motivated behaviors naturally occur. For example, Koch (1956) defined a state (B state) where the person is fully absorbed with the task, has no other need, is elated, energized, and does not feel fatigue. Since then several attempts have been made to develop approaches to personality that can simultaneously accommodate extrinsic and intrinsic motivation and cognition, as well as the role of specific experiential states or patterns (e.g., Buss & Plomin, 1984; Eckblad, 1981; Mahoney, 1984; Rychlak, 1977; Singer, 1985).

The Theory of Flow

One of the most ambitious attempts to integrate motivation, personality, and subjective experience into a unified framework is constituted by the flow theory (Csikszentmihalyi, 1975, 1978, 1982, 1988a, 1990).

This theory attempts to explain the causes of variations in subjective experience and the relationship between the daily dynamic of experience and a large class of seemingly unrelated psychological phenomena such as talent (Csikszentmihalyi, Rathunde, & Whalen, 1993), creative achievement (Csikszentmihalyi, 1988b), and mental health (Massimini, Csikszentmihalyi, & Carli, 1987).

Flow theory represents the person as a system that acts according to three integrated "teleonomies" or motivational systems: the genetic teleonomy, the cultural teleonomy, and the teleonomy of the self. The genetic teleonomy corresponds to seeking pleasures and goals that are genetically programmed in the person's organism, such as eating and being healthy or sexually satisfied. The cultural teleonomy consists of seeking and maintaining social and economical success. The teleonomy of the self leads to reorganization and growth in the order and complexity of consciousness, which the person experiences as enjoyable and rewarding even in the absence of concomitant genetic and cultural prompt or reinforcement. The flow theory postulates that the optimal conditions for subjective experience (flow in consciousness) occur when the person is primarily driven by the teleonomy of the self.

The teleonomy of the self is characterized as follows. The quality of subjective experience is a function of two subjective variables (Massimini et al., 1987): (a) the perceived challenges (or intrinsic demands) of an activity; and (b) the perceived skills (or self-perceived capacity to meet demands). When both challenges and skills are perceived to be low, the person experiences apathy and the overall quality of subjective experience is the lowest. If challenges are perceived to be greater than skills, the person experiences anxiety. If skills are perceived to be greater than challenges, the person experiences boredom. In situations characterized by the simultaneous presence of high perceived challenges and high perceived skills, the person experiences flow in consciousness and the overall quality of subjective experience is the highest. Flow is defined as a psychological state in which the person feels simultaneously cognitively efficient, motivated, and happy.

The theory postulates that people seek conditions under which they can experience flow by means of a twofold dynamic (Csikszentmihalyi & Massimini, 1985): (a) If the perceived skills are less than the perceived challenges in a given task, then a person experiences anxiety and will try to balance by learning new skills; and (b) if the perceived skills are greater than the perceived challenges, then a person experiences boredom and will try to balance by seeking more challenging activities.

The teleonomy of the self is an endless maximization process toward higher and higher levels of perceived challenges and skills, wherein the target of the optimization is experience itself, rather than a predefined result.

The simultaneous growth of action opportunities and action capabilities implies an increase of the content and organization of consciousness, which in turn leads to objectively more efficient and focused attention. The intrapsychic correlate of this growth in cognitive efficiency is the feeling of being "absorbed" by the activity. This perception is postulated to be intrinsically rewarding and to yield the maximum hedonic tone. In this sense, the flow theory views hedonic tone as a function of cognitive efficiency. Furthermore, the theory states that the successful search for flow is likely to enhance not only the overall quality of experience but also mental health, objective performance, talent, and creativity. In this sense, flow theory views achievement as a function of how the person's optimization of experience is systematically successful in the course of his or her life.

The theory states that the systematic search for flow is more likely to occur and is more intense and complex in structured activities, such as the arts, sports, and certain occupations. This is because only in structured activities is the person allowed substantial control over his or her skills and challenges by learning new skills and by selecting the type and difficulty of the task. The theory further states that there are strong individual differences in how the three teleonomies are hierarchically ordered: Some individuals develop an "autotelic personality" in which the dynamic of flow heavily characterizes their choices and experiences. A fundamental implication is that those persons who are mostly driven by the teleonomy of the self are more likely to become talented and creative.

The theory of flow is related to other theories of experience and motivation. The definition of the flow state is similar to the definitions advanced by Koch (1956) and Mahoney (1984), as is the emphasis on pursuit of flow in subjective experience. The theory of self-determination (Deci & Ryan, 1985a, 1985b) also links the pursuit of flow with intrinsic motivation. The idea that hedonic tone can be viewed as a function of stimulus complexity is present in the theory of Berlyne (e.g., 1960, 1970) and in the subsequent scheme-based theories of motivation (e.g., Eckblad, 1981). Finally, there is a clear relationship between the concept of "autotelic personality" and the concept of "hardiness" (Kobasa, Maddi, & Kahn, 1982; Kobasa, Maddi, & Puccetti, 1982), which identi-

fies a stable commitment to confront problems, to cope with complexity, and even to search for more difficult situations. McAdams (1990) has placed the flow theory in the context of the several theories developed in the past 20 years.

It is useful to point out two distinctive features of flow theory. The first is that the function for the optimization of experience is defined on two entirely subjective experiential variables: challenges and skills—the metric of “stimulus complexity” is entirely based on the person’s perception of the activity and of himself or herself. Thus, theories like those of Berlyne (1960, 1970) and Eckblad (1981), which are based on objective definitions of task difficulty and competence, cannot be directly translated into the language of flow theory. From this point of view, flow theory can be labeled as phenomenological and existential (e.g., McAdams, 1990).

The second unique feature of flow theory is that the hypothesized function of experience does not have a maximum or equilibrium point. Both Berlyne (1960, 1970) and Eckblad (1981) propose a limit for the expression of intrinsic motivation wherein individuals seek tasks that are neither too complex nor too simple. Flow theory, instead, assumes no limit: The “logic” of the teleonomy of the self is a continuous search for greater complexity. Hence, flow theory can be looked at as a “strong” conception of intrinsic motivation.

The Measurement of Subjective Experience

It is universally recognized that the optimal approach to studying experience is the collection of repeated measures obtained on the same individuals at different times or occasions (e.g., Stone, Kessler, & Haythornthwaite, 1991; Wheeler & Reis, 1991). These data are structurally very complex and thus there is the need of searching for adequate, more sophisticated statistical treatments of the data (e.g., West & Hepworth, 1991).

There are many methods for measuring experience (Pervin, 1985). Wheeler and Reis (1991) have proposed a convenient classification depending on the timing of measurement: (a) “interval contingent,” where participants are required to respond at regular intervals (as in the end-of-day diaries), (b) “signal contingent,” where participants respond when signaled (for example, by a pager), and (c) “event contingent,” where participants respond when a specified event occurs. One of the most popular signal-contingent methods for evaluating ex-

perience in natural environments is the Experience Sampling Method (ESM) (Csikszentmihalyi & Larson, 1987; Csikszentmihalyi, Larson, & Prescott, 1977; deVries, 1992; Wong & Csikszentmihalyi, 1991). The ESM consists of repeated self-reports in response to randomly generated electronic signals transmitted to pagers that respondents wear for a week. The ESM seems to provide a valid and representative record of what people do, think, and feel, and it allows intrapersonal as well as interpersonal comparisons in the quality of daily experience across times and situations.

The Daily Experience of Adolescents

Csikszentmihalyi and Larson (1984) administered the ESM to a sample of high-school students in the Chicago area and analyzed how the quality of subjective experience varied across four social contexts of activity: life in the family, with friends, in solitude, and at school. These contexts yielded quite different patterns of average values of the ESM Likert-like measures. Life in the family was commonly associated with feeling happy but aggravated by lack of concentration and involvement; companionship with friends bore a higher happiness and involvement but still a low concentration; solitude led to deterioration of experience with respect to happiness and involvement but to higher concentration; and school life yielded unhappiness but high concentration and average involvement. Csikszentmihalyi and Larson interpreted these patterns in terms of the flow theory, that is, by examining the types of activities that are carried on within each context in respect to the levels of challenges and skills that they involve.

Similar experiential patterns were later observed among Italian (Carli, Delle Fave, & Massimini, 1988) and South Korean adolescents (Won, 1989). In particular, these and subsequent studies converged in showing that, when in class or doing homework, children and adolescents report an overall negative experience and a specific lack of intrinsic motivation. At the same time, and somewhat paradoxically, the quality of experience during school activities appears to be of crucial importance for predicting academic success. For example, Larson (1988) found that the level of enjoyment during the writing of a "junior theme" correlated with the teacher's rating of the theme.

The enjoyment of school activities seems to play a relevant role among talented students. For example, Nakamura (1988) studied experiential differences among talented math students and found that

achievement correlated positively with the enjoyment of schoolwork and negatively with the level of anxiety experienced during schoolwork. For a group of talented high-school students, Wong and Csikszentmihalyi (1991) found that intrinsic motivation while studying predicted the difficulty level of the courses that the students took during a follow-up of 4 years. Recently, Csikszentmihalyi et al. (1993) found that the experiential patterns during adolescence predicted whether a talented student remained engaged in his or her talent area 4 years later.

These and other studies have three implications for our investigation. First, they suggest that the analysis of experience has to be conducted within selected social contexts with particular focus on the scholastic environment. Second, the analysis of experience should simultaneously focus on enjoyment as well as on the perceptions of motivation and cognitive efficiency. Third, it is probably easier to observe the autotelic optimization of subjective experience among talented adolescents than among nontalented ones.

Validation of Flow Theory

Flow theory makes many predictions that are potentially testable. One of them, simple and of crucial importance, is that the balance of challenges and skills has a positive effect on experience. This hypothesis was tested for the first time by applying the ESM to a sample of Italian adolescents (Massimini et al., 1987). The experience of flow was operationalized as a response in which a person rates his/her levels of challenges and skills as being above his/her own personal averages computed over the entire week of the ESM administration. When adolescents were in the operationalized flow condition they felt significantly happier, more satisfied, stronger, more motivated, and more cognitively efficient than in any other condition. The results were explained as being due to the simultaneous presence of high challenges and high skills, as predicted by the theory.

The work of Massimini et al. (1987), however, failed to rule out simpler interpretations. Given the adopted operationalization of flow, the same results could have been obtained if the measures of subjective experience were positively associated with either challenges, with skills, or with both variables independently. Thus, the claim that the balance of challenges and skills is the key factor for the optimization of everyday life experience was not exhaustively supported by their analysis.

Goals and Plan of This Study

In this article we discuss the results of the testing of two simple but crucial hypotheses put forth by the flow theory: (a) There is a positive association linking daily subjective experience with the perceived levels of challenges and skills; and (b) the balance of challenges and skills enhances the quality of experience (or, conversely, the imbalance of challenges and skills is negatively associated with experience). Our objective is to estimate the separate effects of challenges and skills on experience, and, in addition, to assess the effect of the balance between challenges and skills as a distinct component of subjective experience. The disentangling of these three effects will clarify whether or not the results obtained by Massimini et al. (1987) admit a simpler explanation than that given by flow theory. The purpose of our investigation is to validate the two most basic and general predictions made by the theory.

It is important to point out that these predictions are not unique to flow theory and could also be derived from other standpoints such as those of Koch (1956), Berlyne (1960, 1970), Eckblad (1981), and Deci and Ryan (1985a, 1985b). Yet, as pointed out before, only flow theory postulates that the "space" for optimization of experience is itself purely experiential, private, and constructed over "perceived" challenges and skills. Thus, given this distinctive feature, the result of the test of the two hypotheses contributes more powerfully to the validation of flow theory than to the validation of the other theories.

The study population is a sample of talented high-school students that has already been the object of several investigations that utilized the ESM (e.g., Csikszentmihalyi et al., 1993; Wong & Csikszentmihalyi, 1991). These adolescents are ideal for our study because, if the theory is correct, they are likely to exhibit the postulated experiential pattern. At the same, if they do not exhibit the postulated pattern, the violation of the flow model can be expected to be greater among nontalented adolescents.

The objective is pursued within the framework of regression analysis. The analysis focuses on four dimensions of subjective experience: concentration, wish to do the activity, involvement, and happiness, which serve as dependent variables. Perceived challenges and skills in the activity serve as predictors. Both the dependent variables and the predictors are measured by the ESM. The hypothesis that the balance of challenges and skills has a positive effect on the quality of experience is regarded to imply a specific type of interaction between challenges and

skills. This interaction is parameterized and tested in the form of the linear absolute difference of challenges and skills. Random coefficients regression modeling (Bryk & Raudenbush, 1992; Goldstein, 1987) is used to allow for individual differences on how strongly challenges and skills may affect the quality of experience, as well as for matching the structural complexity of the ESM data. The regression model is fitted to both a random sample of everyday life experiences and subsamples corresponding to the four different social contexts of activity: being in school, with family members, with friends, and alone. The comparison of parameter estimates obtained across contexts of activity will provide further insight into the applicability of the theoretical model. The complexity and relative novelty of the employed analytical technique require an extensive elaboration of the Method section.

METHOD

Participants

The data were collected at the University of Chicago within a large project for studying talented adolescents throughout the academic year 1984/1985 (Csikszentmihalyi et al., 1993). The participants were students from two prestigious suburban high schools in Chicago. The teachers nominated 371 students as showing the greatest talent in one or more of the following subject areas: mathematics, science, music, athletics, and the arts. Two hundred eight (56%) students volunteered and completed the study. The participants were 92 males and 116 females, and their age ranged from 14 to 17 years.

Sampling of Daily Experiences

The adolescents' activity and subjective experience were studied using the ESM (Csikszentmihalyi et al., 1977; Csikszentmihalyi & Larson, 1987; de Vries, 1992; Wong & Csikszentmihalyi, 1991). The ESM consists of administering the same questionnaire to participants repeatedly at random times during their daily activities. The ESM questionnaire used in this study (Csikszentmihalyi & Larson, 1987) contained both categorical and scaled items. The categorical items aimed to reconstruct the activity and its social context. The scaled items were designed to measure the intensity of associated subjective feelings. Each participant was given an electronic pager activated by radio signal and a block of identical questionnaires to carry for 7 consecutive days. The transmitter signaled about eight times a day at random time intervals. After every pager call, the participants had to fill out a questionnaire. On average, the questionnaire took 2 minutes to be completed.

Measures

A subset of items from the ESM questionnaire was utilized in the analysis. The selected dependent variables were (a) concentration (CONC), (b) wish to do the activity (WISH), (c) involvement (INVO), and (d) happiness (HAPPY). The variables were measured based on the following items from the ESM questionnaire:

- CONC: "How well were you concentrating?"
- WISH: "Do you wish you had been doing something else?"
- INVO: Likert scale detached-involved
- HAPPY: Likert scale sad-happy

CONC and WISH were 10-point scales with values ranging from "not at all" (0) to "very" (9). The responses to INVO and HAPPY were assigned scores ranging from 1 ("not at all") to 7 ("very"). The predictors' perceived challenges (CHALL) and perceived skills (SKILL) were evaluated by the following items:

- CHALL: "Challenges of the activity"
- SKILL: "Your skills in the activity"

Both items were 10-point scales with values ranging from "low" (0) to "high" (9).

Four social contexts of activity were considered in the analysis: being in school (SCHOOL), being with family members (FAMILY), being with friends (FRIENDS), and being alone (ALONE). The four contexts were identified on the basis of the open-ended item "Where were you?" and the closed item "Who were you with?", which yielded the following admissible answers: "alone," "mother," "father," "sister(s) or brother(s)," "friend(s)," "strangers," and "other." In some instances, participants provided multiple answers (e.g., "mother" and "friends") that, if neglected, could confound the specific effects of different contexts. This potential confounding was avoided by selecting the contexts in such a way that they were mutually exclusive but not exhaustive. The context SCHOOL included all the observations collected when participants were physically in school; if when in school they also reported to be in one or more of the remaining contexts, these observations were considered as belonging to the SCHOOL context and not to the others. The observations obtained when the participants were not in school were attributed to the remaining three contexts and the observations corresponding to overlapping contexts were discarded. Thus, for example, all the observations corresponding to the mixed context "mother" and "friend(s)" were eliminated.

Our selection of variables and social contexts was motivated by the previously obtained results. Several studies conducted on diverse samples (e.g.,

Csikszentmihalyi & Larson, 1984; Larson, 1989a) have systematically indicated that the ESM scales have a four-dimensional structure: affect, activation, cognitive efficiency, and motivation. Our selected dependent variables play a key role in three of these dimensions. HAPPY is the central scale of the dimension of affect, and CONC is the central scale of the dimension of cognitive efficiency. Both INVO and WISH belong to the dimension of motivation, although they appeared to vary somewhat differently across activities and their social contexts (e.g., Csikszentmihalyi & Larson, 1984). The predictors CHALL and SKILL were used by Csikszentmihalyi and LeFevre (1989) on a sample of blue-collar workers and by Massimini et al. (1987) on a sample of adolescents. Finally, the selected social contexts correspond to the contexts investigated by Csikszentmihalyi and Larson (1984) in their study of adolescents.

Validation studies of the ESM scales are reviewed in Csikszentmihalyi and Larson (1987). Although it is recognized that there is no conclusive method to validate the individual, beeper-level measures (e.g., Larson, 1989b), these studies have provided three types of supporting evidence: (a) The measures vary in expected ways across activities and their social contexts, and furthermore, the profiles are remarkably parallel across different cultures and different samples of both adults and adolescents; (b) the distributions of the individual measures differ predictably between groups of different populations; and (c) individual measures such as the means and standard deviations of the ESM scores, obtained either in the entire week of the study or in specific contexts, correlate with independent measures of similar constructs. Larson (1989b) reported that average self-assessed affect correlated significantly with Kovacs's (1983) Children's Depression Inventory scores ($r = -.34$), teachers' ratings of mood level ($r = .28$), and the students' GPA ($r = .21$) in a sample of adolescents.

The reliability of the ESM scales has been investigated by comparing the average self-assessed scores obtained in the first and second halves of the week of the study. In a sample of adolescents, Larson (1989b) reports split-half correlations of .66 for HAPPY, .62 for WISH, and .72 for a scale measuring "paying attention." Test-retest correlations over a 2-year lag in adolescents (Freeman, Larson, & Csikszentmihalyi, 1986) also were fair: .77 for HAPPY, .52 for INVO, and .45 for CONC.

In this application of the ESM, a total of 8,711 questionnaires (coded experience samples) were completed (about 75% of the pager signals). The first completed questionnaire of every participant was considered as a practice trial and thus eliminated. An additional 692 questionnaires were discarded because either they were grossly incomplete or they were completed more than 30 minutes after the signal. Thus, the basic data set for the analysis consisted of 7,811 questionnaires (about 59% of the pager signals). For the analysis of each dependent measure, listwise elimination of missing data was employed, leaving 7,547 observations for CONC, 7,533 for WISH, 7,444 for INVO, and

Table 1
Observed Means and Standard Deviations (in Parentheses) of the
Predictors and Dependent Variables for the Entire Domain
of Experience and across Contexts of Activity

	Entire domain	Context of activity			
		SCHOOL	FAMILY	FRIENDS	ALONE
Number of observations	7,811	3,913	1,129	710	1,568
CHALL ^a	3.5 (3.0)	4.0 (3.0)	2.3 (2.6)	3.3 (2.9)	3.2 (3.1)
SKILL ^a	6.2 (2.6)	6.1 (2.5)	6.3 (2.8)	6.2 (2.7)	6.3 (2.7)
CONC ^a	4.4 (2.6)	4.5 (2.6)	4.1 (2.5)	4.2 (2.5)	4.4 (2.6)
WISH ^a	4.1 (3.3)	3.5 (3.1)	4.7 (3.3)	5.7 (3.2)	4.4 (3.3)
INVO ^b	4.6 (1.6)	4.6 (1.5)	4.5 (1.5)	5.3 (1.4)	4.1 (1.6)
HAPPY ^b	4.9 (1.3)	4.8 (1.3)	4.8 (1.3)	5.4 (1.4)	4.6 (1.4)

Note. CHALL = challenge; CONC = concentration; WISH = wish to do the activity; INVO = involvement; HAPPY = happiness.

a. Range of the scale = 0–9; midpoint = 4.5.

b. Range of the scale = 1–7; midpoint = 4.

7,491 for HAPPY. The nonresponse rate was of similar magnitude to that of previous administrations of the ESM. Two validation studies conducted on different samples of adolescents (Csikszentmihalyi & Larson, 1987; Larson, 1989b) pointed out that the three major causes of questionnaire nonresponse are (a) pager malfunctions, (b) leaving the pager or questionnaires at home, and (c) being in places such as a church or a swimming pool where it would be inconvenient to respond. In general, these causes do not seem to constitute a serious threat to the validity of the data analysis (Csikszentmihalyi & Larson, 1987; Larson, 1989b).

Split-half reliabilities based on averages obtained in the first 3 days of the ESM administration versus those obtained in the last 4 days were fair: .57 for CHALL, .84 for SKILL, .67 for CONC, .55 for WISH, .67 for INVO, and .71 for HAPPY. Since comparable numbers of participants began the ESM administration on different days of the week, the above figures are probably uninfluenced by a weekday versus weekend bias.

Table 1 shows the observed means and standard deviations of the variables

Table 2
Observed Correlations among the Predictors and Dependent Variables

	CHALL	SKILL	CONC	WISH	INVO	HAPPY
CHALL	1.000					
SKILL	.055	1.000				
CONC	.508	.165	1.000			
WISH	-.111	.085	.068	1.000		
INVO	.178	.080	.284	.215	1.000	
HAPPY	.005	.133	.105	.290	.430	1.000

Note. The minimum number of pairwise observations was 7,444. CHALL = challenge; CONC = concentration; WISH = wish to do the activity; INVO = involvement; HAPPY = happiness.

for the entire domain of experience and the four contexts of activity. The number of observations refers to the completed ESM questionnaires over the whole sample of participants. The mean values of the dependent variables across contexts exhibit a pattern similar to that obtained by Csikszentmihalyi and Larson (1984).

Table 2 shows the matrix of correlation coefficients involving all the variables. SKILL and CHALL were uncorrelated. The correlations between the dependent variables were modest. This finding supports the discriminant validity of the scales, with the possible exception of INVO and HAPPY. The same correlation matrix was separately computed for the observations gathered in each social context, and no marked intercontext differences were detected. This datum supports the feasibility of the comparisons involving models fitted to the different social contexts.

Statistical Modeling

The analysis used the following regression model:

$$y = \beta_0 + \beta_1 \text{CHALL} + \beta_2 \text{SKILL} + \beta_3 \text{DIFF}$$

where y is one of the four dependent variables; β_0 is the intercept, that is, the expected value of the dependent variable y when both challenges and skills have the value of zero; CHALL and SKILL are linear predictors; and DIFF is the absolute difference between challenges and skills, i.e., $|\text{CHALL} - \text{SKILL}|$. (This parameterization, as well as a class of alternative ones, was suggested by Richard L. Tate.) The model is fully consistent with the theory if the coefficients β_1 and β_2 are greater than zero and β_3 is smaller than zero. If the

estimates β_1 and β_2 are greater than zero, it would imply that both challenges and skills have a positive linear effect on subjective experience, consistent with the findings of Massimini et al. (1987). This is the formulation of our first research hypothesis. In addition, if the estimate β_3 is smaller than zero, then the model implies that the imbalance of challenges and skills has a negative effect on subjective experience; thus, the term DIFF can be regarded as a simple and adequate parameterization of the concept of balance in flow theory. The requisite that the term DIFF must be negative is the formulation of our second research hypothesis.

The model was fitted separately to the four dependent variables. For each dependent variable, the model was first fitted to the entire set of observations of everyday life experience collapsed over contexts. The model was then separately fitted to the observations gathered in the four different contexts of activity, permitting a separate test of predictions in each context.

Random Coefficients Modeling

The fact that each individual was observed in repeated occasions complicates the estimation of the regression model. The problems can be illustrated in connection with two traditional methods of analysis. The simplest approach is to fit a single standard least squares regression model to all the observations performed on all the participants, so that one set of average (population) estimates of the regression coefficients for the entire group of participants is obtained. However, this method fails to recognize that the measures collected on the same individual are statistically dependent. The consequence of ignoring the within-subject correlation of responses is that, although the point estimates of the regression coefficients will be unbiased, their standard errors will be negatively biased (i.e., too small) (e.g., Bryk & Raudenbush, 1992; Goldstein 1987), and significance tests will too often lead to rejection of the null hypothesis. The magnitude of the problem depends on how high the within-subject correlations are (e.g., Goldstein, 1987).

The natural alternative is a two-stage analytic approach. In the first stage, one performs separate least squares regressions on the data of each participant. In the second stage, the population estimates of the regression coefficients are obtained by averaging over the estimated individual regression coefficients. This approach, commonly referred to as "means-as-outcomes regression" (e.g., Bryk & Raudenbush, 1992), provides population estimates and useful information on the extent to which the regression coefficients differ among the individuals in the group. While addressing the problem of within-subject correlation, the approach is limited (Bryk & Raudenbush, 1992): While the estimates of the regression coefficients are unbiased, their standard errors are excessively large, leading to underrejection of the null hypothesis. These limitations are aggravated by two features of the data: the presence of an unequal

number of observations across participants (a common problem if some observations are missing), and the presence of unequally spaced predictor values across participants (e.g., for the j th observation a predictor assumes different values for different participants). Our data set has the above characteristics and thus cannot be analyzed by this method.¹

Fortunately, the limitations of the traditional approaches can be overcome to a large extent by adopting the recently devised statistical methods known as "multilevel modeling" (ML) (Goldstein, 1987) and "hierarchical linear modeling" (HLM) (Bryk & Raudenbush, 1992). These approaches share the same underlying logic and notation and produce estimates that, although obtained via different algorithms, are virtually identical for large data sets like ours. As explained by Goldstein (1987), Bryk and Raudenbush (1992), and Tate and Hokanson (1993) in the context of personality psychology, the analysis of repeated observations can be validly and efficiently performed by a two-level regression model. This approach provides unbiased and maximally efficient estimates of the population regression coefficients, as well as estimates of the extent of individual response differences in relation to the predictors. Most important for our application, the two-level modeling retains the properties of unbiasedness and maximal efficiency in the presence of unequal numbers of repeated observations and unequally spaced predictor values (Bryk & Raudenbush, 1992; Tate & Hokanson, 1993).

A two-level regression model is specified by a Level-1 model (person model), a Level-2 model (population model), and the combined model. For the analysis of repeated measures, the single observations (e.g., the ESM measures collected in correspondence to one pager call) constitute Level 1, and the participants on whom the observations are collected constitute Level 2. The Level-1 model is a within-subject regression equation. The Level-2 model is a set of equations where the parameters of the Level-1 model are expressed as functions of systematic population characteristics ("fixed effects") plus person-specific characteristics that are randomly distributed over the population ("random effects"). The combined model is obtained simply by substituting the parameters in the Level-1 model with their corresponding Level-2 expressions. The combined model integrates all the within-subject models into one regression equation that expresses both the average effects for the population and the degree of heterogeneity of these effects across the participants in the population.

Level 1 provides a model for the relationship of challenges and skills to subjective experience for an individual subject i and observation j :

1. There are 100 possible pairs of values of CHALL and SKILL. Since the number of observations per participant was considerably fewer, the individual data distributions over the values of CHALL and SKILL presented several missing cells. Both the number and the pattern of missing cells varied markedly between participants.

$$y_{ji} = \pi_{0i} + \pi_{1i}\text{CHALL}_{ji} + \pi_{2i}\text{SKILL}_{ji} + \pi_{3i}\text{DIFF}_{ji} + e_{ji}$$

where y_{ji} is a measure of subjective experience provided by subject i in occasion j (i.e., pager call); CHALL_{ji} is the assessment of challenges provided by subject i in occasion j ; SKILL_{ji} is the assessment of skills provided by subject i in occasion j ; the terms π_{0i} through π_{3i} are the individual regression coefficients (person-specific effects of the predictors), assumed to be constant across occasions; e_{ji} is the error, assumed to be normally distributed with mean 0 and variance σ^2 . Conceptually, there is a Level-1 equation for each participant. The participants are allowed to have different numbers of observations (occasions); the occasions can be unequally spaced over time; the timing of the occasions may differ across participants; for every occasion j different participants are allowed to have different predictor values; and, finally, the individual regression coefficients may vary across participants. Note that these individual regression coefficients are not the overall β_0 through β_3 in the original model.

The Level-2 model involves the β_0 through β_3 coefficients discussed above. These coefficients are treated as fixed over the whole population of potential participants. The definition of the Level-2 model requires a number of assumptions concerning the population, the selection of participants, and the Level-1 model. First, the participants are assumed to be randomly sampled from the same population. Second, there is a natural heterogeneity in regression coefficients across the individuals of the population. Third, the individual regression coefficients have a multivariate normal distribution in the population with mean, variances, and covariances to be estimated. Fourth, the residuals of the Level-1 model are distributed independently of the individual regression coefficients. Finally, the residuals of the Level-1 model are uncorrelated among themselves.

Based on the above assumptions, we formulate the Level-2 model, which shows how each coefficient π for the Level-1 equation is related to the corresponding fixed population coefficient β :

$$\pi_{0i} = \beta_0 + r_{0i}$$

$$\pi_{1i} = \beta_1 + r_{1i}$$

$$\pi_{2i} = \beta_2 + r_{2i}$$

$$\pi_{3i} = \beta_3 + r_{3i}$$

The terms π_{0i} through π_{3i} are the individual regression coefficients of subject i . The terms β_0 through β_3 are the average regression coefficients over the entire study population (from which the group of participants was sampled). These coefficients are called fixed effects. The terms r_{0i} through r_{3i} are the unique increments to the fixed effects characteristic of subject i ; they covary over the population of participants following a multivariate normal distribution with

means equal to zero and variances and covariances to be estimated. The variances and covariances of these increments are called random effects. Thus, the Level-2 model expresses the individual regression coefficients as a function of systematic features of the population plus randomly distributed individual difference components.

The random effects are ordered into a covariance matrix. For our model, the covariance matrix has the form:

	CONST	CHALL	SKILL	DIFF
CONST	$\begin{bmatrix} \text{CONST/CONST} & & & \\ \text{CONST/CHALL} & \text{CHALL/CHALL} & & \\ \text{CONST/SKILL} & \text{CHALL/SKILL} & \text{SKILL/SKILL} & \\ \text{CONST/DIFF} & \text{CHALL/DIFF} & \text{SKILL/DIFF} & \text{DIFF/DIFF} \end{bmatrix}$			
CHALL				
SKILL				
DIFF				

where the diagonal components represent the variances and the below-diagonal components the covariances. The term CONST stands for intercept. Each variance on the diagonal of this matrix measures the variability across participants in their individual regression coefficients for one predictor. These variances thus measure individual differences, and will be approximately zero if there are no individual differences among participants in how a predictor relates to an outcome. It is possible to test statistically whether these variances differ from zero, as a test of significance of individual differences. For example, DIFF/DIFF is the variance of the individual regression coefficients π_{3i} of the predictor DIFF (the imbalance); if DIFF/DIFF differs from zero, there are individual differences among participants in how DIFF affects subjective experience. In addition, and most important for our application, dividing the estimated fixed effect β_3 of the predictor DIFF by the square root of the estimated random effect DIFF/DIFF provides us with a useful standardized statistic. Since we assume that the individual coefficients are normally distributed around β_3 , this standardized statistic is normally distributed and its estimated value, when referred to the cumulative standardized normal distribution function, identifies the percentage of the population for which DIFF has a negative (or positive) effect on experience.

The covariance matrix is also used to compute tests of significance of the population effects β_0 through β_3 by which the hypotheses are operationalized. In particular, the covariance matrix enters the estimation of asymptotic standard errors of the fixed effects. From the assumption of normality of the individual regression coefficients it follows that the ratio of each estimated fixed effect by its estimated standard error is a z statistic. Hence, the significance of the fixed effects can easily be evaluated with reference to the cumulative standardized normal distribution function.²

2. It is important to stress that the square roots of the diagonal components (variances) of the covariance matrix are not the standard errors of the fixed effects. Derivations of such standard errors are presented in Goldstein (1986) and in the technical appendix

The combined model can now be expressed in three parenthetical parts:

$$y_{ji} = (\beta_0 + \beta_1 \text{CHALL}_{ji} + \beta_2 \text{SKILL}_{ji} + \beta_3 \text{DIFF}_{ji}) + (r_{0i} + r_{1i} \text{CHALL}_{ji} + r_{2i} \text{SKILL}_{ji} + r_{3i} \text{DIFF}_{ji}) + (e_{ji})$$

Any individual response on one questionnaire is a function of the Level-2 average regression coefficients (fixed effects), the Level-1 individual difference terms, and a Level-1 error component. The extent of the individual differences in the population is expressed by the Level-2 variances and covariances of the individual difference terms (random effects). The extent of the error for the population is expressed by the Level-1 residual within-subject error variance. Given this model, the statistical objective is to estimate the fixed effects, their corresponding random effects, and the within-subject error variance. Furthermore, hypothesis testing requires estimation of the standard errors of each one of the above statistics. Our model belongs to the subtype of two-level models called "random coefficients models" or "unconditional models" (Bryk & Raudenbush, 1992).

Computations

All the fixed and random effects are estimated conjointly, in a multivariate fashion, maximizing statistical efficiency. The estimation is iterative and comes to completion when the estimates become stable at the specified level of precision. We estimated the models by using the program ML3 (Prosser, Rasbash, & Goldstein, 1991), which is based on the theory of multilevel modeling developed by Goldstein (1987). The method of estimation was Iterative Generalized Least Squares (IGLS) (Goldstein, 1986). The criterion of convergence for each estimate was that the updated estimate and its value in the previous iteration differed by less than 1%.

Significance Testing

As indicated above, a z test is used for the significance of each population effect β_0 through β_3 . One-sided hypothesis testing was performed, at $\alpha = .05$, following specific predictions from flow theory. The significance of the random effects (variances reflecting individual differences) was evaluated by the two-tailed normal test with the critical value of 1.96 for the z statistic to reject the null hypothesis.

of Bryk and Raudenbush (1992). These sources also explain how to obtain standard errors of the random effects, and that the ratio of each estimated random effect by its estimated standard error is a z statistic. Thus the significance of the random effects can also be evaluated by the normal test.

Treatment of the Predictors

We kept the predictors CHALL and SKILL in raw score form for the analysis. Thus, the intercept is the expected value of the dependent variable when both CHALL and SKILL are equal to zero. The subject-specific intercept can be interpreted as the subject-specific fixing of the lower bound of the scale. Hence, allowing participants to have their own intercept controls for individual differences in mean subjective experiential levels. Furthermore, the estimation of the fixed effects CHALL, SKILL, and DIFF is performed adjusting for the individual differences of intercept. We believe that this method of controlling for individual differences in scaling is more appropriate and statistically efficient than standardizing predictors because it uses all the information in the data in a simultaneous, multivariate fashion. Another reason for preferring raw scores is that the interpretation of the models is simpler. Once the estimates of the fixed effects are obtained, one may compute expected values of the dependent variable for each pair of values of CHALL and SKILL: These triples of values (outcome, CHALL, SKILL) have the same interpretation for every participant.

Response Surfaces

Once the fixed effects are estimated, the model identifies one expected value of the dependent variable for each combination of values of the predictors. With two predictors, the set of expected values of the dependent variable can be represented as a response surface, where the predictors are the two orthogonal axes on the plane, and the dependent variable is the vertical axis. Visual inspection of the response surface provides insightful information on how to interpret the model.

Table 3 shows three grids of expected values of a hypothetical dependent variable based on three different sets of hypothetical estimates of the fixed effects. In each grid, the number shown in correspondence of each pair of values of CHALL and SKILL is the value of the dependent variable that is expected on the basis of the given parameter estimates. The estimates of Example (a) are fully consistent with the theoretical model; the estimates of Example (b) are inconsistent because the imbalance of challenges and skills has no effect ($\beta_3 = 0$); the estimates of Example (c) are inconsistent because challenges have a negative linear effect ($\beta_1 = -.03$). Since the violations (b) and (c) were both encountered in the analysis, it is useful to illustrate their implications.

Focusing on Example (a), we see that for each value of CHALL (row), the dependent variable takes higher values as SKILL increases. Analogously, for each value of SKILL (column), the dependent variable takes higher values as CHALL increases. Furthermore, for each "balanced" observation (CHALL

Table 3
Expected Values of a Hypothetical Dependent Variable Based on
Three Sets of Hypothetical Estimates of the Fixed Effects

CHALL	SKILL									
	0	1	2	3	4	5	6	7	8	9
Example (a): $\beta_0 = 1, \beta_1 = .4, \beta_2 = .4, \beta_3 = -.05$										
9	4.15	4.60	5.05	5.50	5.95	6.40	6.85	7.30	7.75	8.20
8	3.80	4.25	4.70	5.15	5.60	6.05	6.50	6.95	7.40	7.75
7	3.45	3.90	4.35	4.80	5.25	5.70	6.15	6.60	6.95	7.30
6	3.10	3.55	4.00	4.45	4.90	5.35	5.80	6.15	6.50	6.85
5	2.75	3.20	3.65	4.10	4.55	5.00	5.35	5.70	6.05	6.40
4	2.40	2.85	3.30	3.75	4.20	4.55	4.90	5.25	5.60	5.95
3	2.05	2.50	2.95	3.40	3.75	4.10	4.45	4.80	5.15	5.50
2	1.70	2.15	2.60	2.95	3.30	3.65	4.00	4.35	4.70	5.05
1	1.35	1.80	2.15	2.50	2.85	3.20	3.55	3.90	4.25	4.60
0	1.00	1.35	1.70	2.05	2.40	2.75	3.10	3.45	3.80	4.15
Example (b): $\beta_0 = 1, \beta_1 = .4, \beta_2 = .4, \beta_3 = 0$										
9	4.6	5.0	5.4	5.8	6.2	6.6	7.0	7.4	7.8	8.2
8	4.2	4.6	5.0	5.4	5.8	6.2	6.6	7.0	7.4	7.8
7	3.8	4.2	4.6	5.0	5.4	5.8	6.2	6.6	7.0	7.4
6	3.4	3.8	4.2	4.6	5.0	5.4	5.8	6.2	6.6	7.0
5	3.0	3.4	3.8	4.2	4.6	5.0	5.4	5.8	6.2	6.6
4	2.6	3.0	3.4	3.8	4.2	4.6	5.0	5.4	5.8	6.2
3	2.2	2.6	3.0	3.4	3.8	4.2	4.6	5.0	5.4	5.8
2	1.8	2.2	2.6	3.0	3.4	3.8	4.2	4.6	5.0	5.4
1	1.4	1.8	2.2	2.6	3.0	3.4	3.8	4.2	4.6	5.0
0	1.0	1.4	1.8	2.2	2.6	3.0	3.4	3.8	4.2	4.6
Example (c): $\beta_0 = 1, \beta_1 = -.03, \beta_2 = .5, \beta_3 = -.05$										
9	0.28	0.83	1.38	1.93	2.48	3.03	3.58	4.13	4.68	5.23
8	0.36	0.91	1.46	2.01	2.56	3.11	3.66	4.21	4.76	5.21
7	0.44	0.99	1.54	2.09	2.64	3.19	3.74	4.29	4.74	5.19
6	0.52	1.07	1.62	2.17	2.72	3.27	3.82	4.27	4.72	5.17
5	0.60	1.15	1.70	2.25	2.80	3.35	3.80	4.25	4.70	5.15
4	0.68	1.23	1.78	2.33	2.88	3.33	3.78	4.23	4.68	5.13
3	0.76	1.31	1.86	2.41	2.86	3.31	3.76	4.21	4.66	5.11
2	0.84	1.39	1.94	2.39	2.84	3.29	3.74	4.19	4.64	5.09
1	0.92	1.47	1.92	2.37	2.82	3.27	3.72	4.17	4.62	5.07
0	1.00	1.45	1.90	2.35	2.80	3.25	3.70	4.15	4.60	5.05

Note. CHALL = challenge.

= X , $SKILL = X$), the dependent variable takes higher values than for all the “imbalanced” observations ($CHALL = X + I$, $SKILL = X - I$) and $CHALL = X - I$, $SKILL = X + I$), where I is an integer number. For example, the observation ($CHALL = 5$, $SKILL = 5$) yields higher expected value than the observations ($CHALL = 6$, $SKILL = 4$) and ($CHALL = 4$, $SKILL = 6$). This illustrates the meaning of the expression “the imbalance of challenges and skills has a negative effect.”

Focusing on Example (b), we see that, analogously to the previous case, the dependent variable grows both in respect to $CHALL$ and in respect to $SKILL$. The crucial difference is that in Example (b) each balanced observation $CHALL = SKILL$ has identical expected value to both the observations $CHALL + I = SKILL - I$ and the observations $CHALL - I = SKILL + I$, where I is an integer number. This illustrates the meaning of the expression “the imbalance of challenges and skills has no effect.”

Focusing on Example (c), we see that for each value of $CHALL$ (row), the dependent variable takes higher values as $SKILL$ increases. This feature of the response surface is the same as for Examples (a) and (b). Yet, unlike Examples (a) and (b), it is not true that for each value of $SKILL$ (column), the dependent variable takes higher values as $CHALL$ increases. The behavior of the response surface is more complex. The slope of the surface in respect to $CHALL$ varies depending on the level of $SKILL$: For each value of $SKILL$, the slope is positive for $CHALL \leq SKILL$ and negative for $CHALL > SKILL$; for $SKILL = 9$ the slope is uniformly positive. This illustrates the meaning of the expression “the negative effect of $CHALL$ is counterbalanced by the negative effect of the term $DIFF$.” In this connection, it is important to note that if the fixed effect $DIFF$ is smaller in absolute value than the fixed effect $CHALL$, then the slope of the surface in respect to $CHALL$ will simply become less negative as $SKILL$ increases. This illustrates the meaning of the expression “the negative effect of $CHALL$ is attenuated by the negative effect of the term $DIFF$.” On the other hand, if the fixed effect $DIFF$ is greater in absolute value than the fixed effect $CHALL$, then the slope of the surface in respect to $CHALL$ will behave as in our Example (c).³

3. Three shortcomings of our analytic approach should be pointed out. First, in multi-level modeling computational problems often arise; here, in a few instances, we failed to achieve convergence of the estimates because the variance of the term $DIFF$ was too small. In those cases the problem was obviated by treating the predictor $DIFF$ as a fixed effect; that is, the variance and all the covariance components of the term $DIFF$ were set to zero. The second limitation is that intercontext differences could have been more efficiently analyzed by regression models with dummy variables to code contexts, rather than by our more qualitative inspection approach. Our analytical choice was dictated by the impossibility of estimating the computationally more complex models with dummy variables. The third limitation is that no attempt was made to control for the possible autocorrelation of responses. Autocorrelation applies to the Level-1 residuals, and it can be either positive or negative. Although methods for modeling

RESULTS

Concentration

Table 4 shows the results of the fit for the dependent variable concentration (CONC). Looking at the model for the entire domain of experience, we see that all the fixed effects are significant in the postulated direction. Turning attention to the models for the contexts of activity, we see that the estimates of the predictors CHALL and SKILL are significantly positive for every context. The term DIFF is significantly negative only for the contexts SCHOOL and ALONE. These findings suggest that the negative effect of the imbalance of challenges and skills on concentration occurs only during school activities and when the adolescents are alone.

Wish to Do the Activity

Table 5 shows the results of the fit for the dependent variable wish to do the activity (WISH). In respect to the entire domain of experience, we see that the fixed effects SKILL and DIFF are both significant in the postulated direction, while CHALL is significantly negative, contrary to expectations. Challenges alone appear to increase the wish of doing something else. Inspecting the response surface (not shown) we found that the negative effect of challenges is attenuated as skills increase. In regard to the contexts of activity, we see that the fixed effect CHALL is significantly negative for every context but FRIENDS, and SKILL is significantly positive for every context. The term DIFF is significantly negative for every context except ALONE. Inspecting the response surfaces (not shown), we found that the negative effect of CHALL is attenuated as SKILL increases in the contexts SCHOOL, FAMILY, and ALONE. For the context FRIENDS, the slope of the response surface in respect to CHALL is uniformly positive for $SKILL = 9$.

Involvement

Table 6 shows the results of the fit for the dependent variable involvement (INVO). In the model for the entire domain of experience, all the fixed effects are significant with the hypothesized signs. Focusing

autocorrelation in multilevel models are currently being introduced (Goldstein, Healy, & Rasbash, 1994), the formidable complexity of the issues in connection to the ESM data has led us to postpone such a modeling to a future article.

Table 4

Concentration (CONC): Estimated Fixed Effects (with Standard Errors in Parentheses), Random Effects, and Within-Subject Error Variance

	Entire domain	Social context of activity			
		SCHOOL	FAMILY	FRIENDS	ALONE
Participants	208	208	195	174	205
Observations	7,547	3,801	1,101	692	1,538
Fixed effects					
CONST	2.395* (.101)	1.890* (.132)	3.136* (.221)	2.357* (.243)	2.578* (.182)
CHALL	.373* (.018)	.422* (.020)	.356* (.042)	.348* (.045)	.353* (.032)
SKILL	.165* (.018)	.204* (.023)	.082* (.045)	.119* (.045)	.170* (.037)
DIFF	-.079* (.018)	-.079* (.024)	-.068 (.046)	.004 (.044)	-.082* (.038)
Random effects (variances only)					
CONST/CONST	.818*	.899*	2.492*	1.065	1.177*
CHALL/CHALL	.031*	.026*	.045*	.078*	.029
SKILL/SKILL	.018*	.028*	.000	.021	.025
DIFF/DIFF	.017*	.029*	f	.000	.035
Within-subject error variance	3.617*	3.444*	3.248*	3.546*	3.483*

Note. CONST = intercept; CHALL = challenge; DIFF = absolute difference between challenge and skill; f = effect set to fixed due to problems in achieving convergence.

* $p < .05$, one-sided for the fixed effects.

on the models for the contexts of activity, we see that the fixed effects CHALL and SKILL are both significantly positive for every context. The term DIFF is significantly smaller than zero only for the context SCHOOL. These findings suggest that the negative effect of the imbalance of challenges and skills on involvement occurs only during school activities.

Happiness

Table 7 shows the results of the fit for the dependent variable happiness (HAPPY). Regarding the model for the entire domain of experience, we see that the fixed effects SKILL and DIFF are significant in the pos-

Table 5
Wish to Do the Activity (WISH): Estimated Fixed Effects
 (with Standard Errors in Parentheses), Random Effects,
 and Within-Subject Error Variance

	Entire domain	Social context of activity			
		SCHOOL	FAMILY	FRIENDS	ALONE
Participants	208	208	195	173	205
Observations	7,533	3,790	1,096	693	1,540
Fixed effects					
CONST	3.459*	2.743*	4.721*	5.543*	3.996*
	(.174)	(.211)	(.310)	(.423)	(.289)
CHALL	-.198*	-.167*	-.269*	-.052	-.241*
	(.024)	(.026)	(.067)	(.067)	(.048)
SKILL	.261*	.290*	.215*	.147*	.234*
	(.027)	(.029)	(.071)	(.074)	(.059)
DIFF	-.096*	-.137*	-.167*	-.130*	-.079
	(.024)	(.027)	(.073)	(.069)	(.057)
Random effects					
(variances only)					
CONST/CONST	2.633*	2.863*	3.301*	6.242*	3.277*
CHALL/CHALL	.044*	.038*	.084*	.113*	.044
SKILL/SKILL	.046*	.038*	.000	.117*	.097
DIFF/DIFF	.017	.024*	.000	.000	.035
Within-subject					
error variance	8.517*	7.298*	7.629*	6.335*	7.842*

Note. CONST = intercept; CHALL = challenge; DIFF = absolute difference between challenge and skill.

* $p < .05$, one-sided for the fixed effects.

tulated direction. Yet the fixed effect SKILL is significantly negative, contrary to expectations. Inspecting the response surface (not shown), we found that the negative effect of CHALL is counterbalanced by the negative effect of DIFF: The slope of the surface in respect to CHALL is uniformly positive for SKILL = 9. Looking at the estimates across contexts of activity, we notice that the fixed effect CHALL is significantly negative for every context except for FRIENDS. The effect SKILL is significantly positive for every context. The term DIFF is significantly negative for every context. The inspection of the response surfaces (not shown) revealed that in all four contexts the negative effect of CHALL

Table 6

Involvement (INVO): Estimated Fixed Effects (with Standard Errors in Parentheses), Random Effects, and Within-Subject Error Variance

	Entire domain	Social context of activity			
		SCHOOL	FAMILY	FRIENDS	ALONE
Participants	207	207	193	172	203
Observations	7,444	3,749	1,088	682	1,516
Fixed effects					
CONST	4.021* (.080)	3.948* (.100)	4.231* (.127)	4.684* (.167)	3.741* (.132)
CHALL	.061* (.009)	.052* (.012)	.060* (.027)	.090* (.025)	.077* (.021)
SKILL	.077* (.012)	.097* (.015)	.058* (.031)	.067* (.026)	.049* (.025)
DIFF	-.037* (.010)	-.042* (.015)	-.042 (.031)	-.015 (.026)	-.029 (.024)
Random effects (variances only)					
CONST/CONST	.593*	.638*	.487*	.810*	.636*
CHALL/CHALL	.003*	.002	.006	.010	.010*
SKILL/SKILL	.001	.007	.008	.000	.005
DIFF/DIFF	.001	.008*	f	.000	.000
Within-subject error variance	1.814*	1.747*	1.329*	1.259*	1.754*

Note. CONST = intercept; CHALL = challenge; DIFF = absolute difference between challenge and skill; f = effect set to fixed due to problems in achieving convergence.

* $p < .05$, one-sided for the fixed effects.

is counterbalanced by the negative effect of DIFF: For SKILL = 9, the slope of all the response surfaces in respect to challenges is uniformly positive.

Individual Differences

As shown in Tables 4 through 7, several random effects (variances) achieved significance. This indicates the presence of individual differences and justifies a posteriori our choice of the analytical method. For our application it is interesting to focus on the term DIFF/DIFF, which is the variance of the balance effect across participants. In the models

Table 7
Happiness (HAPPY): Estimated Fixed Effects (with Standard Errors in Parentheses), Random Effects, and Within-Subject Error Variance

	Entire domain	Social context of activity			
		SCHOOL	FAMILY	FRIENDS	ALONE
Participants	208	208	194	174	204
Observations	7,491	3,769	1,094	687	1,527
Fixed effects					
CONST	4.556*	4.486*	4.518*	5.237*	4.257*
	(.071)	(.087)	(.133)	(.167)	(.128)
CHALL	-.036*	-.044*	-.053*	-.026	-.032*
	(.009)	(.011)	(.024)	(.034)	(.018)
SKILL	.093*	.105*	.113*	.081*	.108*
	(.011)	(.012)	(.028)	(.033)	(.021)
DIFF	-.044*	-.045*	-.060*	-.073*	-.045*
	(.010)	(.011)	(.027)	(.034)	(.020)
Random effects					
(variances only)					
CONST/CONST	.524*	.588*	.870*	.995*	.842*
CHALL/CHALL	.006*	.007*	.006	.044*	.010*
SKILL/SKILL	.007*	.003	.011*	.005	.007
DIFF/DIFF	.003*	.002	.000	.014	f
Within-subject error variance	1.231*	1.124*	1.020*	1.262*	1.260*

Note. CONST = intercept; CHALL = challenge; DIFF = absolute difference between challenge and skill; f = effect set to fixed due to problems in achieving convergence.

* $p < .05$, one-sided.

on the entire domain of experience, this term achieves significance for CONC and HAPPY but not for WISH and INVO. This finding suggests that the negative effect of the imbalance of challenges and skills is virtually the same size for all participants (i.e., it is uniform) for wish and involvement; on the other hand, the size of the effect differs across participants (i.e., it is heterogeneous) for concentration and happiness. By examining the z statistics (ratios of the fixed effect β_3 of DIFF by the square root of its random effect DIFF/DIFF), we estimate that the imbalance of challenges and skills has a negative effect on more than 72.8% of the subjects in the population for CONC, 76.9% for WISH, 87.9% for INVO, and 78.9% for HAPPY. Furthermore, looking at the

size and significance of the term DIFF/DIFF across contexts, it appears that the variance of the DIFF effect is mostly expressed in SCHOOL, and it is virtually null in the other contexts. This finding suggests that for all the dependent variables, the negative effect of the imbalance is virtually uniform across participants in all contexts but SCHOOL.

It is worth focusing on the random effect CHALL/CHALL to further qualify the violation of the theoretical model with regard to the dependent variables WISH and HAPPY. For WISH, the random effect is significant for the entire domain of experience and every context but ALONE. By examining the z statistics (ratios of β_1 by the square root of CHALL/CHALL), we estimate that challenges have a negative effect on 80.4% of subjects in SCHOOL, 82.3% in FAMILY, 56.1% in FRIENDS, and 87.5% for ALONE. For HAPPY, the random effect is significant for the entire domain of experience and every context but FAMILY. We estimate that challenges have a negative effect on 70.1% of subjects in SCHOOL, 75.3% in FAMILY, 54.9% in FRIENDS, and 62.6% for ALONE. These findings suggest that for both wish and happiness the observed violation of the theoretical model is least severe when the participants are with friends, where approximately 45% of participants behave according to expectations.

Finally, it is interesting to notice that the random effect CONST/CONST was significant for all dependent variables and across all contexts. This indicates that there were large and consistent individual differences in scale location. Insofar as the individual differences in scaling were accounted for by the random intercept, the estimates of the fixed effects CHALL, SKILL, and DIFF should be reasonably valid and accurate.

Differences of Intercepts across Contexts

As shown in Tables 4 through 7, the most evident differences across contexts concerned the fixed effect CONST. For all dependent variables, the point estimates in the context SCHOOL are lower than in both the FAMILY and FRIENDS contexts. For all dependent variables, the estimates in the context ALONE are lower than in the context FAMILY and, with the exception of concentration, are lower than in the context FRIENDS. By computing the 95% confidence intervals of each estimate we can identify those pairs for which the confidence intervals do not overlap and, thus, whose point estimates differ significantly. For concentration, the estimate for SCHOOL is significantly smaller than

for both FAMILY and ALONE. For wish, the estimate for SCHOOL is significantly smaller than for all the other contexts; furthermore, the estimate for ALONE is smaller than for FRIENDS. For involvement, the estimates for SCHOOL and ALONE are both significantly smaller than for FRIENDS. For happiness, the estimate for FRIENDS is significantly greater than for all the other contexts. These findings indicate that when both challenges and skills are null the expected values of the dependent variables differ across contexts; in general, SCHOOL and ALONE tend to yield the lowest values, and FAMILY and FRIENDS the highest values.

DISCUSSION

In this article we investigated two hypotheses: (a) that there is a positive association linking the quality of daily subjective experience with the perceived levels of challenges and skills in the activity and (b) that the balance of challenges and skills enhances the quality of experience. In the analysis on the entire domain of experience, the first hypothesis was confirmed for the variables concentration and involvement. Yet the hypothesis was disconfirmed for the variables wish and happiness, in that challenges had a negative relationship to both variables. For wish, the negative effect of challenges becomes smaller as skills increase. For happiness, the effect of challenges becomes positive for highest values of skills. On the other hand, the second hypothesis turned out to be fully confirmed for all four dimensions of experience, in that the quality of experience was negatively related to the imbalance of challenges and skills.

In the analysis on the four social contexts of activity, the negative effect of the imbalance of challenges and skills was detected for 10 of the 16 models. In particular, school activities yielded the effect for all four dimensions of experience. In family life the effect was detected for wish and happiness but not for concentration and involvement. In being with friends we found the effect for wish and happiness but not for concentration and involvement. And when the adolescents were in solitude we detected the effect for concentration and happiness but not for wish and involvement. These findings suggest that the balance of challenges and skills has a positive effect in some contexts and little or no effect in others. Furthermore, even within the same context the balance of challenges and skills may have a positive effect on one dimension of experience and no effect on others. Thus, on the whole the

analysis confirms the theoretical model but, at the same time, identifies some circumstances in which the model may not apply.

The failures to detect the negative effect of the imbalance may also be due to lack of statistical power. There were more observations in school than in any other context. This could be a reason why the effect of the imbalance was systematically detected in that context, and only at times in the others. Given that it is still problematic to evaluate the statistical power of multilevel models (e.g., Tate & Hokanson, 1993), the issue of whether the detected intercontext differences are real can be addressed only by repeating the analysis on different samples of participants.

Regarding the effect of the imbalance of challenges and skills, the major difference was between the school context, on the one hand, and the family and friends contexts, on the other hand. Only school activities yielded a significantly negative effect of the imbalance on concentration and involvement. Keeping in mind the above reservations on the power of tests, these findings suggest that the flow model may be more applicable to social contexts and activities where achievement plays a dominant role than to those ones where social interaction is the main issue. If school is able to provide the student with simultaneously high levels of challenges and skills, then this context is ideal for the student's optimization of concentration and involvement.

It should be stressed that the differences found between the school and family contexts represent an additional qualification to the differences found previously by Csikszentmihalyi and Larson (1984). In their study, the differences concerned the *average* values of the several dimensions of experience. In this study, we replicated this observation. In addition, we found differences of *models* of optimization of experience. The findings imply that the observed differences of average values across contexts can be explained in part by intercontext differences of the effects of challenges, skills, and their relative balance. In particular, the pattern of low average values of concentration and involvement observed in the family context, which Csikszentmihalyi and Larson labeled "familiar aggravations" of experience, is also due to the fact that the balance of challenges and skills appears to play a limited role in family life. We could not fully explain the intercontext differences of average values, but we have pointed out one explanatory factor that is theoretically important.

The greatest differences among the parameter estimates of the four contexts of activity concerned the intercept. The heterogeneity of the

intercept across contexts indicates that the expected quality of experience when both challenges and skills are null differs from context to context. This in turn implies that the differences of mean levels of subjective experience that we observed across contexts are not merely due to the effects of challenges, skills, and their relative balance. Put differently, it appears that the different contexts "produce" different mean levels of concentration, wish, involvement, and happiness. This production can be explained only in part by the flow model as we operationalized it in this article. Therefore, it seems that challenges and skills are not the only factors affecting the quality of these four dimensions of experience.

By inspecting the estimated response surface one identifies the pair of values of challenges and skills for which subjective experience is expected to be maximal. This is not a test of the hypothesis that subjective experience is at a peak when challenges and skills are highest, but rather an implication of the model conditional on the model being valid. For every context, the dimensions concentration, involvement, and happiness reach their maximal expected values when challenges and skills are both highest. The dimension wish behaves identically in the context being with friends, while in the other three contexts it reaches its peak for highest skills and lowest challenges. Thus, with the exception of wish for these three contexts, these findings confirm Massimini and coworkers' (1987) observation that subjective experience peaks when challenges and skills are both greater than average.

The issue that motivated this investigation was whether the observation by Massimini et al. (1987) could be simply due to subjective experience being positively associated either with challenges, with skills, or with both independent of each other. For those dimensions and contexts in which we detected a negative effect of the imbalance of challenges and skills, we can now conclude that the explanation given by Massimini and coworkers was correct. In those 10 cases, the fact that experience was at a peak when challenges and skills are highest was also due to the positive effect of challenges and skills being in relative balance with each other.

Albeit our analysis largely confirmed flow theory, it also pointed out some departures from the ideal simplicity of the theoretical model. These departures concerned differences of regression coefficients across dimensions of experience as well as across social contexts of activity. These heterogeneities call for a modification and extension of flow theory.

In particular, the different behavior of the four dimensions of experience implies that their simultaneous optimization may require trade-offs. In some cases, given a starting pair of values of challenges and skills, to increase one of the two factors of a given amount improves one dimension of experience and, at the same time, decreases the quality of another dimension. For example, based on our estimated models for the context of being alone, if skills are highest and challenges lowest, to increase challenges (i.e., to reduce the imbalance of challenges and skills) leads to higher concentration but lower wish to engage in the activity. The principles of these trade-offs need to be conceptualized and incorporated into flow theory. In particular, it would be important to understand what makes an adolescent actually go for high-challenge/high-skill states, given that the route to those states may imply a loss of happiness and wish at least in some stages of the process. Here, recent research suggests potential explanatory factors. If the problem is faced in terms of personality, "hardiness" (Kobasa, Maddi, & Kahn, 1982; Kobasa, Maddi, & Puccetti, 1982) and "self-determination" (Deci & Ryan, 1985a, 1985b) may be the relevant constructs. From a more cognitive perspective, the response could be found in terms of goals (especially long-term ones), expectations (e.g., Cantor et al., 1991), and schemes (e.g., Eckblad, 1981).

The different behavior of each dimension of experience across contexts of activity is open to two main interpretations. First, the behavior may differ in two contexts because the challenges and skills specific to each context differ in nature. This implies that challenges and skills are aggregate concepts, and both of them can be thought of as labeling classes of specific types of challenges and skills. Second, the behavior of a dimension of experience may differ in two contexts because one or more additional explanatory variables, beyond challenges and skills, are differently distributed in the two contexts. The first interpretation keeps the two-predictor structure but seeks a more detailed specification and measurement of the two predictors. The second interpretation requires that flow theory be expanded beyond the two-predictor structure. Both research paths should be investigated theoretically and empirically in future work.

The suspected violations of the flow model should be interpreted in relation to the study population. It is relevant to ask if the different behavior of the four measures of experience are peculiar to talented adolescents or can be generalized to all adolescents. Based on flow theory, we would expect nontalented adolescents to exhibit stronger violations.

This is because we would expect them to be more driven by the genetic and cultural teleonomies. In turn, these teleonomies should lead to an experiential pattern that elevates happiness and wish to do the activity rather than concentration and involvement. Based on our models, this implies that challenges would have a stronger negative effect, and that the balance/imbalance of challenges and skills would have a smaller effect. From the assumption that talent is related to the power of the teleonomy of the self, we would expect this to happen especially in the school context.

It is difficult to predict if the findings would generalize to talented adults. It could be that talented adults have a unique model for the simultaneous optimization of all dimensions of subjective experience which equally applies to all social contexts. If this is the case, the somewhat conflicting pattern that emerged for this sample would be a temporary phenomenon in the life of adolescents that may disappear as a part of a developmental process. On the other hand, it could also be that for talented adults the balance of challenges and skills plays a relevant role in the context of their talent activities and a less important role in social contexts where achievement is not the main issue.

The answer to these questions requires the development of a broad and articulate empirical investigation. An overview of the methodological limitations of this study may help the planning of further efforts. These limitations concern the measures, the missing observations, the mathematical model, the statistical model, and the incompleteness of models.

For the measures, it would be useful to develop an ESM questionnaire that includes a set of scales targeted to the evaluation of challenges and skills. This would improve the assessment of the predictors by constructing a measurement model and would offer a greater power to interpret intercontext differences of parameter estimates by constructing a taxonomy of different types of challenges and skills. Furthermore, it would be useful to obtain a greater number of observations per participant. This would increase the power of testing and allow researchers to examine differences between more specific contexts of activity.

As in previous applications of the ESM, the nonresponse rate was relatively high. This raises the problem of whether nonresponse induced a bias in the analysis and, if so, in which direction. As we mentioned before, there is evidence supporting the hypothesis that nonresponse to the ESM is mostly due to chance. At the same time, there is preliminary evidence (Prosser, 1991) that the estimation via multilevel modeling,

following a listwise elimination of missing observations, is rather satisfactory in the presence of randomly missing observations. Yet in future applications of the ESM it would be advisable to conduct a validation study specifically targeted to the assessment of performance of the multilevel or hierarchical linear modeling.

With regard to the mathematical modeling, we are aware that the function adopted in this article is not the only reasonable translation of the theoretical statement and that many other functions could have been entertained. Our choice was the result of a trade-off between appropriateness of the translation, simplicity of interpretation, and information content of the data. In this connection, we emphasize that more complex models may not be testable using the present data; furthermore, the precision of the data has to be improved in order to allow conclusive comparisons between different functions.

For the statistical models, we believe that the multilevel or hierarchical linear modeling is to date the best approach to the analysis of the ESM data. Here we failed to fully exploit this analytic technique. For example, we did not test for intercontext differences by indicator variables, and we did not control for the serial correlation of errors. These limitations, dictated by the structural complexity of the data and the insufficient number of observations, should be overcome in future research.

Finally, our modeling neglected the role of personality characteristics, goals, and expectations. This was justifiable in so far as we had strong *a priori* beliefs based on flow theory, and we had a narrowly selected study population. Yet it would be interesting to investigate whether the inclusion of appropriate personality and cognitive factors in the models might reduce, and thus explain, the marked individual differences of parameter estimates that we detected in this study.

In conclusion, this article challenges the scientists studying the optimization of subjective experience to entertain research in three different directions. At the empirical level, the findings on different samples of participants need to be replicated and the possible effect modifications due to personality and cognition need to be investigated. At the theoretical level, flow theory needs to be extended to accommodate the observed departures from the ideal simplicity of the flow model. At the methodological level, the analytical modeling and the measurement methods need to be improved so that more specific research issues can be efficiently tackled.

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